High-Resolution Brain Metabolic Imaging at Ultrahigh Field Using Extended Spatiospectral Encoding and Subspace Modeling

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***Abstract*—*Objective:* To develop a high-resolution magnetic resonance (MR) metabolic imaging method for mapping human brain metabolite distributions at ultrahigh field (7T). *Methods:* In data acquisition, a free-induction-decay (FID) based MR spectroscopic imaging (MRSI) sequence was implemented. To achieve high spatial resolution, the sequence used fast echo-planar spectroscopic imaging (EPSI) trajectories with echo-spacings larger than the Nyquist sampling interval. Using this sequence, 3D MRSI signals at isotropic nominal resolutions of 3.0 mm and 1.8 mm were acquired within scan times of 4.8 and 14.2 minutes, respectively. In data processing, model-based methods integrating subspace learning, spectral modeling, and generalized series modeling were developed to address key challenges, including spectral ghosting, low signal-to-noise ratio, and spectral aliasing. *Results:* The proposed acquisition and processing methods successfully generated high-resolution, high-quality metabolite maps of the human brain at 7T. Experimental results from phantom and in vivo scans validated the proposed method and showed its capability to capture detailed brain metabolite distributions. *Conclusion:* This work demonstrates the feasibility of high-resolution brain metabolic imaging at ultrahigh field using MRSI acquisition sequence and model-based processing methods. *Significance:* By providing high-resolution spatial mapping of brain metabolites within clinically feasible scan times, the proposed method promises to offer a powerful imaging tool for investigating brain metabolism, which is expected to be useful for various brain imaging applications.**

***Index Terms*—Metabolic imaging, MR spectroscopic imaging, spatiospectral encoding, subspace modeling, ultrahigh field.**

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I. Introduction

MR SPECTROSCOPIC imaging (MRSI) is a noninvasive, label-free, metabolic imaging technique that enables spatial mapping of multiple molecular biomarkers to assess brain metabolism under normal and diseased states [1]. By providing unique insights into the biochemical environment of the brain, MRSI has shown significant potential for the diagnosis and monitoring of numerous neurological disorders, such as brain tumor, stroke, epilepsy, and neurodegenerative diseases [2], [3], [4], [5], [6]. However, due to the low concentrations of brain metabolites and the large number of spatiospectral encodings required, traditional MRSI methods have long been limited to poor spatial resolutions and lengthy scan times. These result in strong partial volume effects and heavy burden on patients, which have largely restricted MRSI's practical utility for broader clinical applications [7], [8].

In recent decades, extensive research efforts have been devoted to improving the spatial resolution and imaging speed of MRSI, leading to numerous advancements. On the hardware front, high-field and ultrahigh-field MR systems have been developed, providing substantial signal enhancements and increased spectral resolution [9], [10], [11], which are critical for resolving brain metabolites. From the imaging sequence perspective, various fast trajectories have been proposed to accelerate spatiospectral encoding, including echo-planar [12], [13], spiral [14], [15], concentric rings [16], [17], [18], and rosette trajectories [19], [20]. Compared with conventional

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phase encoding-based MRSI (known as CSI) sequences [21], these fast trajectories simultaneously sample spatial and temporal encodings, resulting in a significantly higher spatial encoding efficiency. Additionally, techniques like parallel imaging and compressed sensing have been employed to enable sparse sampling for further acceleration [22], [23], [24]. In post-processing, a series of advanced methods have been proposed for water and lipid removal, artifact correction, denoising, and quantification, to produce high-quality MRSI outcomes [25], [26], [27], [28], [29]. But given the progress, the spatial and temporal resolutions of the state-of-the-art MRSI methods are still significantly lower than those of structural and functional imaging used in practice.

While ultrahigh field MR is promising to enhance the imaging capabilities of MRSI, it introduces several technical challenges, including stronger B_0/B_1 (static and radiofrequency magnetic field) inhomogeneity, higher energy deposition, and faster signal decay owing to shortened transverse relaxation time [10], [30]. Achieving higher resolution for MRSI at ultrahigh field faces even more difficulties. One major challenge is the limited spatial encoding efficiency, constrained by the spectral bandwidth requirements. Ultrahigh field MR demands a wider spectral bandwidth; for example, to cover most metabolite resonances, the required bandwidth at 7T is more than double that at 3T. Consequently, for spatiospectral trajectories at ultrahigh field, the achievable spatial encoding speed is often limited since a fast spectral encoding speed is required to meet the bandwidth requirement (i.e., Nyquist sampling rate) [31], [32]. Multi-shot temporal interleaving can help increase spectral bandwidth to extend spatial encodings, but at the cost of prolonged scan times and potential spectral ghosting artifacts due to inter-shot inconsistencies [33]. Another major challenge is the signal-to-noise ratio (SNR) needed to support higher resolutions. Achieving high spatial resolutions in three dimensions requires a higher SNR that scales cubically with resolution while SNR gains from increased field strength scale only slightly more than linearly [34], [35]. Spatial smoothing and signal averaging can help improve SNR but at the expense of lower spatial resolutions and extended scan times respectively, which will further restrict MRSI's feasibility for practical applications.

In this work, we propose a novel method integrating ultrafast MRSI acquisition and model-based processing to overcome these challenges and achieve high-resolution brain metabolic imaging at 7T. The proposed method generated metabolite maps at 3.0-mm and 1.8-mm nominal resolutions within 4.8 and 14.2 minutes, respectively. The high-resolution metabolite maps successfully captured metabolic heterogeneity across brain regions, revealing details that were difficult to resolve with conventional low-resolution MRSI.

II. Methods

A. Data Acquisition

To overcome the technical challenges at 7T, the proposed data acquisition sequence leveraged basic features of the SPICE (SPectroscopic Imaging by exploiting spatiospectral CorrElation) sequence [36], [37], [38], [39] and was optimized for high-resolution MRSI at ultrahigh field. As illustrated in Fig. 1(a), the sequence incorporated several features: (1) free-induction-decay (FID)-based acquisition with ultra-short echo time (TE); (2) Ernst-angle excitation; (3) partial water suppression; (4) short repetition time (TR); and (5) special echo planar spectroscopic imaging (EPSI) readout trajectories.

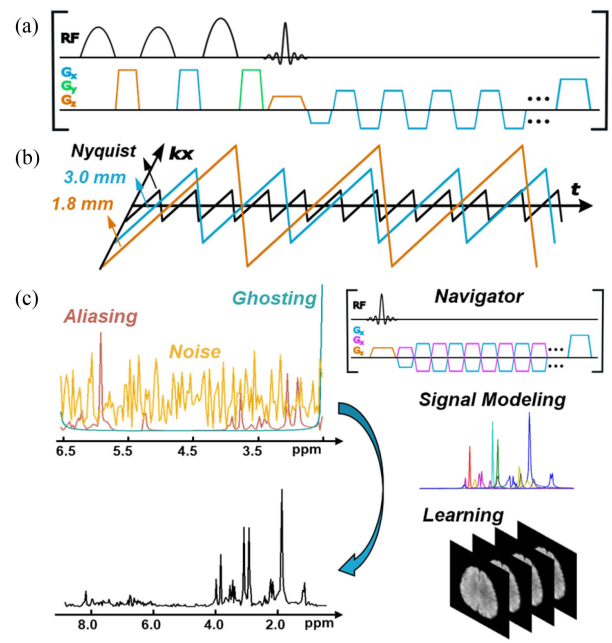

Fig. 1. (a) The proposed sequence uses a free-induction-decay based acquisition and echo-planar spectroscopic imaging trajectory for rapid spatiospectral encoding. (b) The sampling trajectories in (kx, t)-space for Nyquist sampling (monopolar, echo-spacing: 0.40 ms, resolution 10.0 mm), 3.0-mm sampling (bipolar, echo-spacing: 0.52 ms), and 1.8-mm sampling (bipolar, echo-spacing: 0.82 ms). (c) Uncorrected spectra from the acquisition containing spectral ghosting, aliasing and noise. The proposed method integrates navigator signals, signal modeling and learning to address these artifacts and produce high quality spectra. The spectra were produced from a spectroscopy phantom.

FID-based acquisition was selected for its reduced sensitivity to B_1 field inhomogeneity compared with conventional spin-echo MRSI sequences [40]. An ultrashort TE of 1.4 ms was used to reduce signal decay and maximize SNR efficiency. A slice-selective RF pulse centered at -2.4 ppm with an 11 kHz bandwidth was used to reduce chemical shift displacement in the slice direction. Its flip angle was 26° (Ernst angle), optimized for NAA signals. Partial water suppression was achieved using a WET (Water suppression enhanced through T_1 effects) [41] module, which consisted of three Gaussian pulses with flip angles of 76°, 71°, and 120°. The intervals between pulses were 11 ms, 11 ms, and 18 ms and spoiler gradients of 20 mT/m were applied between them, leading to a total duration of approximately 44 ms. This WET module provided a water suppression factor of around 10–30, which effectively attenuated sideband signals while maintaining sufficient water signals for B_0 field estimation and eddy current correction. No spatial saturation pulses or inversion pulses were used, ensuring low RF energy deposition even at 7T. A short repetition time of 150 ms was employed for high acquisition efficiency. Both short TR and TE minimized the effects of faster signal decay at 7T.

To satisfy Nyquist sampling requirement of ^{1}H-MRSI at 7T, a sampling rate of 0.4 ms is usually required, corresponding to 2.5 kHz or 8.4 ppm to cover most metabolite resonances. In contrast, a sampling rate of 0.93 ms is sufficient at 3T to cover an 8.4 ppm range for metabolites. Therefore, the number of spatial encodings per readout is heavily constrained for EPSI trajectories at 7T. For example, with a readout dwell time of 5 μs, a monopolar EPSI readout satisfying Nyquist sampling at 7T (echo-spacing 0.4 ms) could only provide approximately 22 spatial encodings per readout (corresponding to a resolution of about 10.0 mm). In this work (as illustrated in Fig. 1(b)), we used bipolar EPSI readouts and extended the readout length for higher resolutions, even though the echo-spacing violated the Nyquist sampling criterion. More specifically, we used 78 spatial encodings to achieve 3.0 mm nominal resolution (echo-spacing: 0.52 ms) and 120 encodings to achieve 1.8 mm resolution (echo-spacing: 0.82 ms). To maintain fast imaging speed, no multi-shot temporal interleaving was applied. Therefore, the extended echo-spacing led to spectral aliasing due to the limited bandwidth, which needed to be addressed in later processing. Additionally, the bipolar EPSI gradients introduced ghosting due to the odd-even inconsistencies. To correct the spectral ghosting, we collected a set of additional navigator signals at the end of the acquisition. This contained one pair of EPSI readouts at the center of (ky, kz)-plane with opposite readout gradients, as shown in Fig. 1(c). For phase encoding, full Cartesian sampling was used to achieve sufficient spatial encodings. As a result, given a field of view (FOV) of 220×220×72 mm^3, the matrix sizes and scan times for the 3.0-mm and 1.8-mm acquisitions were 78×78×24 in 4.8 minutes, and 120×120×40 in 14.2 minutes, respectively.

The proposed pulse sequences were implemented on 7T MR scanners (MAGNETOM Terra, Siemens Healthcare, Erlangen, Germany) equipped with 1Tx/32Rx head coils (Nova Medical, Massachusetts, USA). To evaluate the feasibility and performance of the proposed method, both phantom and in vivo experiments were carried out. In phantom scans, a uniform spherical spectroscopy phantom with metabolites at physiological concentrations as in the human brain was used (Gold Standard Phantoms, South Yorkshire, United Kingdom). In vivo brain MRSI experiments were conducted on human subjects under approval by the local Institutional Review Board. Written consent form was obtained prior to the study. Twelve volunteers (4 males, 8 females; mean age: 39±15 years) were scanned with the 3.0-mm acquisition and five subjects (5 males; mean age: 31±3 years). were scanned with the 1.8-mm acquisition.

B. Subspace Modeling and Learning

Given such a fast, high-resolution MRSI acquisition at 7T, the measured MRSI signals were corrupted with strong spectral ghosting, heavy noise, and spectral aliasing, as illustrated in Fig. 1(c). Addressing these issues required incorporating strong prior knowledge on the spectral structures of the desired MRSI signals–free from ghosting, noise, and aliasing. In this work, we employed a subspace modeling and learning strategy, which determined spectral subspace structures from physics-informed spectral modeling and training data [36], [42]. These pre-defined subspace structures served as robust, strong priors to effectively overcome the processing challenges.

Specifically, we used a union-of-subspaces model to represent the desired MRSI signals, which assumed that the MRSI signals, including water and metabolite signals, could be represented using low-dimensional subspaces, based on the partial separability [43]:

$$\rho\ (\boldsymbol{x}, t) = \sum_{m=1}^{M} \sum_{l=1}^{L_m} u_{l,m}\,(\boldsymbol{x})\, v_{l,m}\,(t)\ , \tag{1}$$

where $\{v_{l,m}(t)\}$ and $\{u_{l,m}(\boldsymbol{x})\}$ are the temporal basis functions and corresponding spatial coefficients. M and L_m denote the number of molecules and the model rank of mth molecule. To pre-learn the basis functions from the training data (whole set of MRSI data acquired using the same sequence), the MRSI signals at each spatial location can be further expressed as:

$$\rho\ (t) = \sum_{m=1}^{M} c_m \varphi_m\,(t)\, e^{-\frac{t}{T_m} + i2\pi\Delta f_m t} h\,(t)\ , \tag{2}$$

where $\{\varphi_m(t)\}$ is physical basis function created from quantum mechanical simulation containing resonance structures, T_m and Δf_m are the spectral parameters, and c_m is the coefficient. $h(t)$ is a modulation function absorbing lineshape distortions caused by system imperfections and experimental variations, which can be derived from the water signals [36]. The distributions of spectral parameters $\{T_m,\ \Delta f_m\}$ were estimated by fitting training data to this model and used to generate sample signals via (2). Then subspace structures $\{v_{l,m}(t)\}$ were estimated from these synthesized signals. This learning strategy integrated quantum basis and training data to pre-estimate the basis functions of different molecules. The water basis functions provided a ghost-free template, which was essential in reducing spectral aliasing; the metabolite basis functions provided strong spectral priors, facilitating the reconstruction from noisy measurements and overcoming aliasing, as detailed in the following sections.

C. Ghost Correction

A two-step strategy was used to overcome the spectral ghosting. First, the zero-order shift between odd and even EPSI readouts was corrected using the bipolar navigator signals, then a spatially point-by-point correction was performed to remove residual ghosting. More specifically, it is assumed that the main discrepancy between the signals of measured even echoes and inconsistency-free even echoes is linear phases in the readout direction [33]:

$$d_p\ (x, t) = d_0\ (x, t)\, e^{-i2\pi\Delta k x}, \tag{3}$$

where $d_p(x,t)$ and $d_0(x,t)$ are the measured and desired spatiotemporal signals (at even echoes), Δk is the readout shift in k-space to be estimated. In this work, Δk was estimated through linear least-square fitting with $d_p(x,t)$ and $d_0(x,t)$ being the even echoes in the navigator signals with opposite and aligned gradients, respectively. Then this readout shift was applied back to all the even echoes in MRSI data to correct its effects.

For residual ghosting, we assumed the residual inconsistencies were small, which could be captured by generalized series (GS) modeling with low model orders [44], [45]:

$$\rho_i\ (\boldsymbol{x}, t) = \rho_{gf}\ (\boldsymbol{x}, t) \sum_{n\,=\,-N_i/2}^{N_i/2} a_n\,(\boldsymbol{x})\, e^{i2\pi nt/M\Delta T}, \quad (4)$$

where ρ_i are the odd or even echoes of MRSI signals, ρ_{gf} the desired ghost-free MRSI signals, N_i the model orders, ΔT the echo-spacing, and $\{a_n\}$ the model coefficients. To estimate the GS coefficients, we generated the reference water signals ρ_{gf} by projecting measured MRSI signals (odd-even interleaved) onto the ghost-free water subspace learned from training data. With the synthesized reference signals, GS coefficients were estimated by linear least-squares fitting to (4) for odd and even echoes separately:

$$\hat{A} = \arg\min_{A} \|\mathrm{P}_i - G\,(A) \cdot \mathrm{P}_{gf}\|_2^2\,, \quad (5)$$

where P_i, P_{gf}, and A are vector forms of ρ_i, ρ_{gf}, and $\{a_n\}$, respectively. $G(A)$ is the GS function shown in (4). The resulting GS functions $G(\hat{A})$ were divided from $\rho_i(\boldsymbol{x}, t)$ to create the odd and even MRSI signals with ghosting corrected.

D. Reconstruction

Given the high resolutions and short scan times, the acquired MRSI signals had relatively low SNR even at 7T. While the subspace model with learned basis functions has reduced the degrees of freedom to represent the MRSI signals, additional statistical constraints were imposed on the spatial coefficients to improve reconstruction from the noisy measurements [42], [46]. We assumed that the spatial coefficients $\{u_{l,m}\}$ followed certain distributions, $\mathrm{Pr}(\{u_{l,m}\})$. The distributions could be estimated from the spatial coefficients in training data, which were obtained by projecting the MRSI signals onto the learned subspaces. A Gaussian model was used as approximation, with parameters estimated in the maximum likelihood sense [47]. With basis functions and probability distributions established, the reconstruction was formulated as solving the following optimization problem:

$$\hat{U} = \arg\min_{U} \|d - UV\|_2^2 + \lambda\|WU\|_2^2 - \sigma_n^2 \log\left(\mathrm{Pr}\,(U)\right), \quad (6)$$

where U, V, d denote the matrix forms of $\{u_{l,m}(\boldsymbol{x})\}$, $\{v_{l,m}(t)\}$, and noisy MRSI data with nuisance signals removed, σ_n^2 the noise variance, W is the edge-preserved total variation operator, and λ the weighting parameter. With the spatial coefficients determined, the reconstructed MRSI signals were synthesized as $\mathrm{P}_r = \hat{U}\,V$.

E. De-Aliasing

After reconstruction, the MRSI signals still contained spectral aliasing due to the limited bandwidth. To overcome the aliasing issue, we applied a model-based method for temporal interpolation. Specifically, a targeted time grid with half of the echo-spacing $\Delta T/2$ was used, effectively doubling the spectral bandwidth. Denoting the interpolated MRSI signals as $\rho_d(\boldsymbol{x}, t)$, we assumed it can be expressed as:

$$\rho_d\ (\boldsymbol{x}, t) = \sum_{l=1}^{L_d} \gamma_l\,(\boldsymbol{x})\,\psi_l\,(t) \sum_{n\,=\,-N_b/2}^{N_b/2} b_n\,(\boldsymbol{x})\, e^{i4\pi nt/M\Delta T}, \quad (7)$$

where $\{\psi_l(t)\}$ are the metabolite basis functions defined on the interpolated time grid, $\{\gamma_l(\boldsymbol{x})\}$ are the weighting coefficients, and $\{b_n(\boldsymbol{x})\}$ the GS coefficients. The interpolated metabolite basis functions were generated leveraging the fitted spectral parameters during subspace learning while the quantum simulated basis functions were generated using the interpolated time grid for signal synthesis and subspace estimation. With the new basis functions $\{\psi_l(t)\}$, the de-aliasing step was performed for each spatial point by solving the following least-squares optimization problem:

$$\hat{\Gamma},\ \hat{B} = \arg\min_{\Gamma, B} \|\mathrm{P}_r - \Omega_t\,(\Gamma\Psi \cdot G\,(B))\,\|_2^2\,, \quad (8)$$

where Γ, Ψ, B are the vector-form of $\{\gamma_l(\boldsymbol{x})\}$, $\{\psi_l(t)\}$, and $\{b_n(\boldsymbol{x})\}$, respectively. Ω_t is the temporal sampling operator. With the estimated results, interpolated signals were generated using (7).

Other processing steps, such as removal of lipid signals, removal of water signals, correction of B_0 field inhomogeneity, and spectral quantification, were performed using the SPICE processing pipeline as described in the literature [25], [27], [36], [48], [49], [50].

III. Results

Fig. 2 illustrates the experimental results obtained from the spectral phantom using the 1.8-mm MRSI acquisition. The raw spectra, especially those from the edge regions in the readout direction, showed strong spectral ghosting, as shown in Fig. 2(b). After ghost correction using the proposed method (Fig. 2(c)), the spectral ghosting was significantly reduced, making it below the noise level. In Fig. 2(e), the spectra after ghost correction and water removal were heavily corrupted by measurement noise. As can be seen, the noise level was comparable to that of the metabolite signals. Using the proposed reconstruction method (Fig. 2(f)), noise levels were extensively reduced, resulting in minimal noise variations. But the denoised spectra showed spectral aliasing due to its insufficient bandwidth. Fig. 2(h) presents the spectra after temporal interpolation using conventional spline interpolation, whereas they still contained aliased spectral peaks, such as the NAA resonance at 2.0 ppm. In contrast, Fig. 2(i) shows the spectra after de-aliasing using the proposed method, which provided sufficient bandwidth to cover most metabolites.

Fig. 3(a) and (b) display the metabolite ratio maps obtained using 3.0-mm acquisition and 1.8-mm acquisition, respectively. Both sets of ratio maps showed reasonable homogeneity, consistent with the uniform nature of the phantom. The ratio maps from the 1.8-mm acquisition had slightly higher variation, which was caused by the increased noise level in high-resolution scans. Boxplots of the metabolite ratios are presented in Fig. 3(c). The calculated metabolite ratios were 1.249 ± 0.031 (NAA/Cr),

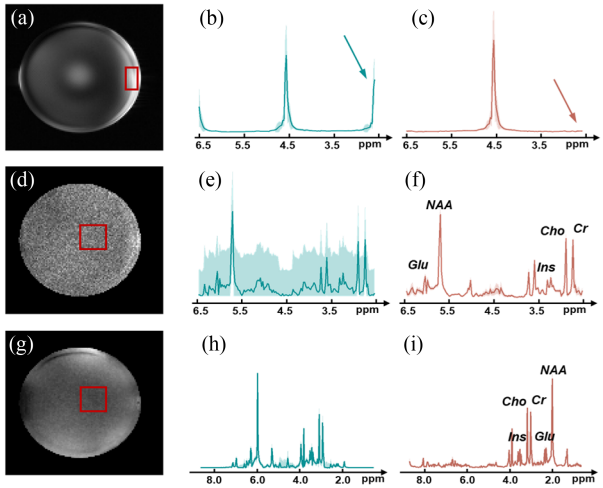

Fig. 2. (a) Spectral integration image of the raw MRSI data. (b) Raw MRSI spectra from the region labelled in (a), (c) ghost corrected MRSI spectra of those in (b), using the proposed method. (d) Spectral integration image of water removed MRSI data. (e) Water removed MRSI spectra from the region labelled in (d). (f) Reconstructed MRSI spectra from those in (e), using the proposed method. (g) Spectral integration image of reconstructed MRSI data. (h) MRSI spectra after spline interpolation, from the region labelled in (g). (i) De-aliased spectra of those in (f), using the proposed method. Magnitude spectra are presented. Solid lines are the mean spectra, and the shading areas indicate the standard deviation. Spectral integration images were calculated as the sum of the squares over the entire MR spectrum at each voxel.

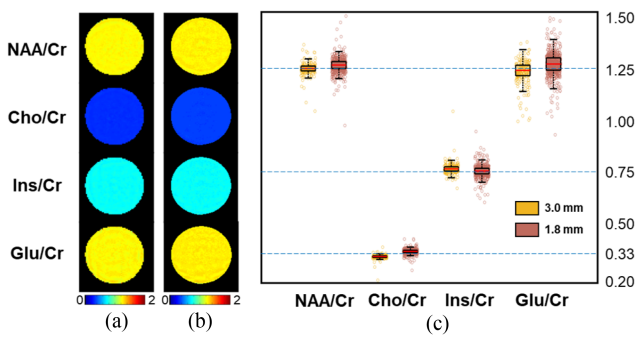

Fig. 3. (a) Metabolite ratio maps from the 3.0-mm acquisition. (b) Metabolite ratio maps from the 1.8-mm acquisition. (c) Boxplots of the metabolite ratios in these two acquisitions. The reference values of this phantom are 1.25, 0.33, 0.75, and 1.25 for NAA/Cr, Cho/Cr, Ins/Cr, and Glu/Cr, respectively.

0.334 ± 0.010 (Cho/Cr), 0.757 ± 0.022 (Ins/Cr), and 1.15 ± 0.051 (Glu/Cr) for 3.0-mm acquisition, and 1.256 ± 0.039 (NAA/Cr), 0.358 ± 0.015 (Cho/Cr), 0.745 ± 0.029 (Ins/Cr), and 1.261 ± 0.060 (Glu/Cr) for 1.8-mm acquisition, which aligned well with the reference values: 1.25 (NAA/Cr), 0.33 (Cho/Cr), 0.75 (Ins/Cr), and 1.25 (Glu/Cr).

Fig. 4 presents a set of in vivo metabolite maps (acquired at 3.0-mm nominal resolution) from four healthy subjects. These metabolite maps exhibited consistently high image quality, with minimal artifacts from spectral ghosting or aliasing, and were resilient to variations resulting from noise. The high resolution achieved by the proposed method allowed for clear visualization of metabolite contrasts between gray and white matters

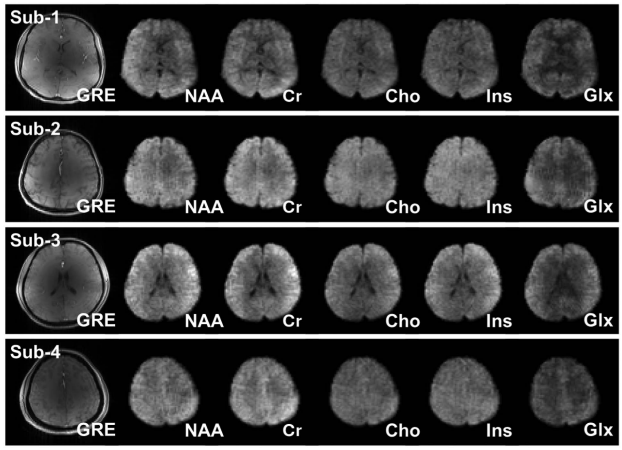

Fig. 4. Representative sets of metabolite maps from four healthy subjects, including NAA, Cr, Cho, Ins, and Glx. The data were acquired using the proposed 3.0-mm acquisition. Excellent quality and clear contrasts were produced for these metabolite maps.

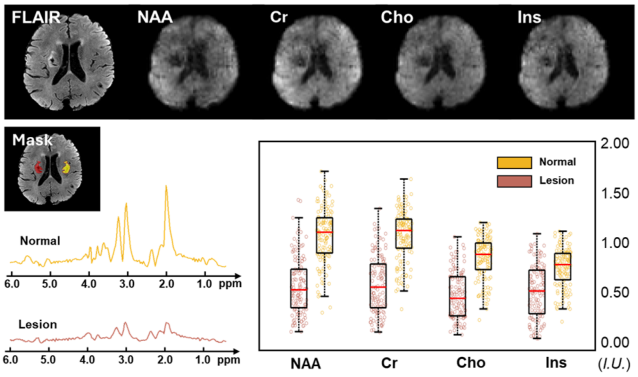

Fig. 5. Metabolite maps acquired from a patient diagnosed with intracerebral hemorrhage, including NAA, Cr, Cho and Ins. The lesion can be observed on the structural FLAIR image. Representative spectra from the lesion and the contralateral normal tissue are displayed. The metabolite concentrations in the lesion and its contralateral normal tissue are compared in the Boxplots. The metabolite levels in the lesion showed a significantly reduced level compared with the contralateral normal tissue.

(especially on NAA, Cr, and Glx maps). Additionally, the absence of metabolites in the brain ventricles was clearly revealed given the high-resolution capabilities. Another set of metabolite maps from a patient diagnosed with intracerebral hemorrhage is displayed in Fig. 5. From both the spatial maps and regional analysis, we could observe the noticeable reduction of metabolite levels in the hemorrhage lesion. The lesion size was roughly 18×12 mm^2 in-plane and its metabolic abnormality was captured by our method, suggesting tissue damage and reduced metabolic activities. These cases demonstrated the resolution capability and sensitivity of the proposed method for potential neurological applications.

Fig. 6 presents a set of metabolite maps acquired at 1.8-mm nominal resolution from one healthy volunteer across several slices. Compared with the 3.0-mm acquisition, the higher resolution provided a more detailed view of the metabolic distributions. More specifically, in the zoomed-in regions of these

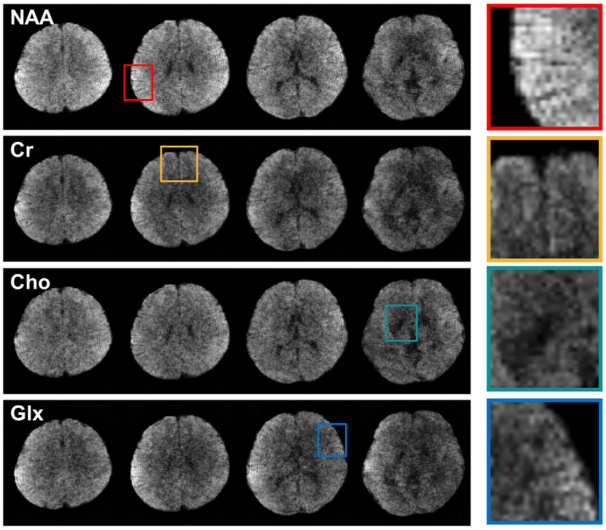

Fig. 6. Representative ultrahigh-resolution metabolite maps (at isotropic 1.8-mm resolution) from a healthy volunteer, including NAA, Cr, Cho, and Glx maps. The zoomed-in regions show more detailed metabolic distributions on cortical layers, longitudinal fissure, deep brain nuclei, etc.

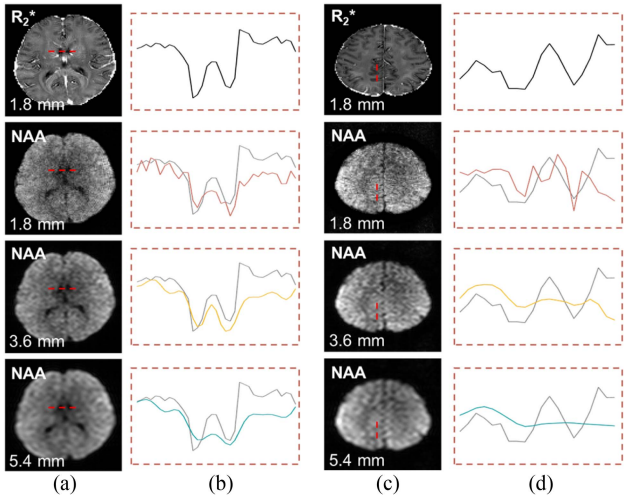

Fig. 7. (a) R_2^* map in 1.8-mm and NAA maps in 1.8-mm, 3.0-mm, and 5.4-mm resolutions from one healthy volunteer. (b) Spatial profiles of R_2^* and NAA distribution under different resolutions in the line segment indicated in (a). (c) R_2^* map and NAA maps from another healthy subject. (d) Spatial profiles of R_2^* and NAA distribution under different resolutions in the line segment indicated in (c).

metabolite maps, finer metabolic heterogeneity was evident in areas such as cortical gray matter layers, the longitudinal fissure, and deep brain nuclei. Notably, the cortical gray matter layers showed higher levels of NAA and Glx compared with the surrounding white matter-dominated tissues, and some structures of gyri and sulci in these layers were observable to a certain extent. The higher NAA and Glx aligned with expectations since these regions have a higher density of neurons, where NAA and Glx are known to be more concentrated. The longitudinal fissure was expected to have minimal metabolites since it is primarily filled with cerebrospinal fluid. Being able to delineate the fissure, which spans only a few millimeters, highlighted the ultrahigh resolution capability of the proposed method in reducing partial volume effect. With lower resolutions, some of the fine structures could be unobservable due to the partial volume effects. Some examples are presented in Fig. 7. The lower- resolution data at 3.6-mm and 5.4-mm resolution were generated from the 1.8-mm data by zeroing out corresponding peripheral k-space data. Fig. 7(a) and (b) examine two adjacent small ventricle structures separated by approximately 10 mm. At 1.8-mm and 3.6-mm resolutions, the separation of these ventricles was clearly visible in the metabolic profiles. However, at a resolution of 5.4 mm, the distinction between the structures became difficult. Fig. 7(c) and (d) showcase results from another subject, focusing on cortical gyri in the parietal lobe. In the NAA map at 1.8-mm resolution, some gyri structures aligned well with features observed in structural imaging. However, as the resolutions were degraded to 3.6-mm and 5.4-mm, these metabolic features were progressively smoothed out and became indistinguishable. These observations showed the necessity of such an ultrahigh resolution to mitigate partial volume effects and resolve such intricate metabolic features.

IV. Discussion

This work integrates ultrahigh field MR, special sequence with extended EPSI spatiospectral readouts, subspace modeling and learning to address key challenges of MRSI techniques at 7T, achieving high-resolution brain metabolic imaging. The 3.0-mm acquisition takes less than 5 minutes, which is comparable to standard MRI scans, making it practical for regular applications. Meanwhile, the 1.8-mm acquisition, which takes a longer time, allows a more detailed view of the brain metabolic heterogeneity, revealing fine features which were not observable using low-resolution methods.

The extended EPSI readouts used in the proposed sequence enabled an acquisition of the entire (kx, t)-plane within a single repetition time, avoiding the need for multi-shot interleaving thus maximizing the acquisition speed. In the tradeoff between spatial and spectral encodings, we sacrificed spectral encoding to satisfy spatial encoding for high-resolution. This choice was based on the relatively predictable nature of MRSI spectral features, as the spectra of brain molecules have fixed resonance structures and limited line-shape variations, and there are only a limited number of brain metabolites detectable by ^{1}H-MRSI. Quantum mechanics simulations could generate physical basis functions with sufficient spectral bandwidths; the pre-learned subspaces determined by integrating these basis functions with training data provided high-quality aliasing-free references, which enabled us to effectively overcome the spectral aliasing caused by sub-sampling. Another key feature that enabled the extended EPSI readouts was the effective correction of spectral ghosting due to eddy currents. Addressing the spectral ghosting allowed the full use of data in both odd and even echoes of the bipolar EPSI readout, gaining encoding efficiency by nearly a factor of two compared with monopolar EPSI readouts. The

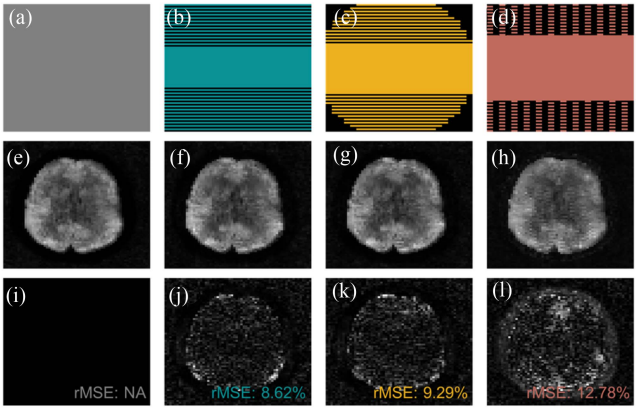

Fig. 8. Illustration of the acceleration using parallel imaging. (a)–(d) Sampling patterns in (ky, kz)-space. The numbers of sampled phase-encoding points in (a)–(d) are 1872, 1224, 1194, 1188, corresponding to scan times (with a TR of 150 ms) of 4.8, 3.0, 3.0, and 3.0 minutes, respectively. (e)–(h) Glx maps reconstructed from the corresponding k-space sampling in (a)–(d). (i)–(l) The difference maps and relative mean square errors (rMSEs) of the reconstruction from different k-space samplings in (a)–(d). Difference maps were displayed in 30% of the dynamic ranges of metabolite maps.

high-order ghosting removal method leveraged the partially suppressed water signals, which still dominated the measured MRSI spectra, to estimate the spatially dependent correcting functions, as water resonances without ghosting were relatively easy to model. This method could be adaptable for other fast MRSI trajectories sensitive to inconsistencies such as spiral-out-in readouts or multi-shot interleaved acquisitions [51]. The reference signals and correcting functions could be estimated either from the accompanying water signals as the proposed method or from additional non-water-suppressed scans.

In current implementation of the sequences, phase encoding directions (ky and kz) were fully sampled to preserve sufficient SNR to support resolution. But further reduction of scan time is possible through integration with other acceleration methods, such as parallel imaging. We tested the feasibility of parallel imaging by retrospectively under-sampling the 3.0-mm MRSI data using several simulated patterns, as illustrated in Fig. 8(b)–(d). The three sampling patterns each had approximately two-thirds of the phase-encoding points of the fully sampled data, corresponding to scan times of about 3 minutes. The under-sampling was applied to the unprocessed k-space raw data, and reconstruction was performed by SENSE (Sensitivity Encoding) reconstruction using sensitivity maps estimated from reference GRE (Gradient Echo) scans [52]. As shown in Fig. 8(j)–(l), the relative errors of these three subsampling patterns were around 10% . Among these, the results from uniform ky under-sampling (Fig. 8(b)) produced the lowest errors relative to the fully sampled reference. The resulting Glx map preserved clear contrast between gray and white matter, demonstrating the feasibility of using parallel imaging to further accelerate the proposed method.

The high-resolution metabolic imaging capabilities achieved using the proposed method are expected to benefit various applications. The 3.0-mm nominal resolution is comparable to the resolution capability of Positron Emission Tomography (PET), and the 1.8-mm nominal resolution is close to typical functional MR imaging. Metabolic imaging in such high resolutions holds great promises in improving delineation of lesions, mapping regional metabolic heterogeneity, revealing subtle metabolic abnormalities in the brain, etc. The ultrahigh field strength played a significant role in supporting such high resolutions. Compared with clinical 3T scanners, 7T systems provide nearly threefold SNR enhancement [35], which can theoretically enable a similar factor of resolution increase. Therefore, the emerging MR systems with even higher field strength such as 10.5T and 11.7T are expected to further improve the resolution capabilities. This feasibility study has demonstrated the capability to capture metabolic abnormalities in hemorrhage lesions (Fig. 5) and reveal metabolic details that closely follow brain structures (Fig. 7). But some more comprehensive studies are necessary to systematically quantify the effective resolution and fully validate the observed metabolic features.

Several potential improvements are envisioned for future development. One is the correction of B_1 homogeneity. The metabolite maps have noticeable spatial shading effects resulting from the strong B_1 inhomogeneity at 7T. Using metabolite ratios can partially correct the B_1 inhomogeneity, but it has limited accuracy and may potentially obscure observable metabolic features. For example, in CSF or lesion areas where metabolite concentrations are similarly low, it may fail to show clear contrast on ratio maps. Therefore, advanced B_1 mapping techniques will be essential to correct the B_1 effects and enable an accurate quantification [53], [54]. Second, in this work, we primarily focused on the major metabolites of the highest concentrations to demonstrate the high-resolution capability. But given the increased spectral resolution at ultrahigh field, it is expected that more molecules could be quantified reliably from the spectra. To achieve reliable mapping of more metabolites, optimization in subspace modeling and learning, improvement in spectral quantification methods, and systematic assessment of accuracy and uncertainty will be needed. Third, the 1.8-mm and 3.0-mm resolutions reported in this work are nominal resolutions. Fully characterizing the true resolution capabilities of the proposed method requires comprehensive studies and evaluations, which will be important directions for future work. Fourth, we are continuing to collect MRSI data through an on-going population study at 7T, which aims to collect more than 200 datasets in total. With more high-quality, high-resolution MRSI data available, including more population priors into the processing using advanced machine learning would further enhance the imaging capabilities of the proposed methods.

V. Conclusion

This work demonstrated the feasibility of high-resolution metabolic imaging of the brain using MRSI acquisition with extended spatiospectral encoding at 7T, combined with processing methods based on subspace modeling and learning. The proposed method offers a new potential for exploring brain metabolism with unprecedented details, which is expected to enhance our understanding of brain function and pathology, providing valuable insights for both research and clinical practice.

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References

[1] S. Posse et al., "MR spectroscopic imaging: Principles and recent advances," *J. Magn. Reson. Imag.*, vol. 37, pp. 1301–1325, 2013, doi: 10.1002/jmri.23945.

[2] W. R. W. Martin, "MR spectroscopy in neurodegenerative disease," *Mol. Imag. Biol.*, vol. 9, pp. 196–203, 2007, doi: 10.1007/s11307-007-0087-2.

[3] M. Castillo, L. Kwock, and S. K. Mukherji, "Clinical applications of proton MR spectroscopy," *Amer. J. Neuroradiol.*, vol. 17, no. 1, pp. 1–15, 1996.

[4] A. Horská and P. B. Barker, "Imaging of brain tumors: MR spectroscopy and metabolic imaging," *Neuroimag. Clin. North America*, vol. 20, pp. 293–310, 2010, doi: 10.1016/j.nic.2010.04.003.

[5] Y. Li et al., "Fast high-resolution metabolic imaging of acute stroke with 3D magnetic resonance spectroscopy," *Brain*, vol. 143, no. 11, pp. 3225–3233, 2021, doi: 10.1093/BRAIN/AWAA264.

[6] Z. Lin et al., "Predicting the onset of ischemic stroke with fast high-resolution 3D MR spectroscopic imaging," *J. Magn. Reson. Imag.*, vol. 58, no. 3, pp. 838–847, 2023, doi: 10.1002/jmri.28596.

[7] D. P. Soares and M. Law, "Magnetic resonance spectroscopy of the brain: Review of metabolites and clinical applications," *Clin. Radiol.*, vol. 64, pp. 12–21, 2009, doi: 10.1016/j.crad.2008.07.002.

[8] G. Öz et al., "Clinical proton MR spectroscopy in central nervous system disorders," *Radiology*, vol. 270, pp. 658–679, 2014, doi: 10.1148/radiol.13130531.

[9] K. Ugurbil, "Magnetic resonance imaging at ultrahigh fields," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 5, pp. 1364–79, May 2014, doi: 10.1109/TBME.2014.2313619.

[10] M. I. Karamat, S. Darvish-Molla, and A. Santos-Diaz, "Opportunities and challenges of 7 Tesla magnetic resonance imaging: A review," 2016, doi: 10.1615/CritRevBiomedEng.2016016365.

[11] M. Lu, W. Chen, and X. H. Zhu, "Field dependence study of in vivo brain 31P MRS up to 16.4T," *NMR Biomed.*, vol. 27, no. 9, pp. 1135–1141, 2014, doi: 10.1002/nbm.3167.

[12] S. Posse et al., "High speed 1H spectroscopic imaging in human brain by echo planar spatial-spectral encoding," *Magn. Reson. Med.*, vol. 33, no. 1, pp. 34–40, 1995, doi: 10.1002/mrm.1910330106.

[13] A. A. Maudsley et al., "Mapping of brain metabolite distributions by volumetric proton MR spectroscopic imaging (MRSI)," *Magn. Reson. Med.*, vol. 61, no. 3, pp. 548–559, 2009, doi: 10.1002/mrm.21875.

[14] E. Adalsteinsson et al., "Volumetric spectroscopic imaging with spiral-based *k*-space trajectories," *Magn. Reson. Med.*, vol. 39, no. 6, pp. 889–898, 1998, doi: 10.1002/mrm.1910390606.

[15] I. Chatnuntawech et al., "Accelerated 1H MRSI using randomly undersampled spiral-based *k*-space trajectories," *Magn. Reson. Med.*, vol. 74, no. 1, pp. 13–24, 2015, doi: 10.1002/mrm.25394.

[16] M. Chiew et al., "Density-weighted concentric rings *k*-space trajectory for 1H magnetic resonance spectroscopic imaging at 7 T," *NMR Biomed.*, vol. 31, no. 1, 2018, Art. no. e3838, doi: 10.1002/nbm.3838.

[17] G. Hangel et al., "High-resolution metabolic imaging of high-grade gliomas using 7T-CRT-FID-MRSI," *Neuroimage, Clin.*, vol. 28, 2020, Art. no. 102433, doi: 10.1016/j.nicl.2020.102433.

[18] J. K. Furuyama, N. E. Wilson, and M. A. Thomas, "Spectroscopic imaging using concentrically circular echo-planar trajectories in vivo," *Magn. Reson. Med.* vol. 67, pp. 1515–1522, 2012, doi: 10.1002/mrm.23184.

[19] S. Z. Mahmud, T. S. Denney, and A. Bashir, "High-resolution proton metabolic mapping of the human brain at 7 T using free induction decay rosette spectroscopic imaging," *NMR Biomed.*, vol. 37, no. 1, 2024, Art. no. e5042, doi: 10.1002/nbm.5042.

[20] C. V. Schirda et al., "Fast 3D rosette spectroscopic imaging of neocortical abnormalities at 3 T: Assessment of spectral quality," *Magn. Reson. Med.*, vol. 79, no. 5, pp. 2470–2480, 2018, doi: 10.1002/mrm.26901.

[21] T. R. Brown, B. M. Kincaid, and K. Ugurbil, "NMR chemical shift imaging in three dimensions," *Proc. Nat. Acad. Sci.*, vol. 79, no. 11, pp. 3523–3526, 1982, doi: 10.1073/pnas.79.11.3523.

[22] W. Bogner, R. Otazo, and A. Henning, "Accelerated MR spectroscopic imaging—A review of current and emerging techniques," *NMR Biomed.*, vol. 34, Art. no. e4314, 2021, doi: 10.1002/nbm.4314.

[23] A. Klauser et al., "Whole-brain high-resolution metabolite mapping with 3D compressed-sensing SENSE low-rank 1H FID-MRSI," *NMR Biomed.*, vol. 35, 2021, Art. no. e4615, doi: 10.1002/nbm.4615.

[24] P. Cao and E. X. Wu, "Accelerating phase-encoded proton MR spectroscopic imaging by compressed sensing," *J. Magn. Reson. Imag.*, vol. 41, no. 2, pp. 487–495, 2015, doi: 10.1002/jmri.24553.

[25] C. Ma et al., "Removal of nuisance signals from limited and sparse 1H MRSI data using a union-of-subspaces model," *Magn. Reson. Med.*, vol. 75, no. 2, pp. 488–497, 2016, doi: 10.1002/mrm.25635.

[26] H. M. Nguyen et al., "Denoising MR spectroscopic imaging data with low-rank approximations," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 1, pp. 78–89, Jan. 2013, doi: 10.1109/TBME.2012.2223466.

[27] Y. Li et al., "A subspace approach to spectral quantification for MR spectroscopic imaging," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 10, pp. 2486–2489, Oct. 2017, doi: 10.1109/TBME.2017.2741922.

[28] D. Hernando et al., "Removal of lipid signal in MRSI using spatial-spectral constraints," in *Proc. 4th IEEE Int. Symp. Biomed. Imag.: From Nano to Macro - Proc.*, 2007, pp. 1360–1363, doi: 10.1109/ISBI.2007.357113.

[29] C. I. Haupt et al., "Removal of lipid artifacts in 1H spectroscopic imaging by data extrapolation," *Magn. Reson. Med.*, vol. 35, no. 5, pp. 678–687, 1996, doi: 10.1002/mrm.1910350509.

[30] G. Barisano et al., "Clinical 7T MRI: Are we there yet? A review about magnetic resonance imaging at ultra-high field," *Br. J. Radiol.*, vol. 92, 2019, Art. no. 20180492, doi: 10.1259/bjr.20180492.

[31] R. Schmidt, A. Seginer, and A. Tal, "Combining multiband slice selection with consistent k-t-space EPSI for accelerated spectral imaging," *Magn. Reson. Med.*, vol. 82, no. 3, pp. 867–876, 2019, doi: 10.1002/mrm.27767.

[32] E. Coello et al., "High-resolution echo-planar spectroscopic imaging at ultra-high field," *NMR Biomed.*, vol. 31, no. 11, 2018, Art. no. e3950, doi: 10.1002/nbm.3950.

[33] A. Seginer et al., "Artifact suppression in readout-segmented consistent K-t space EPSI (RS-COKE) for fast 1H spectroscopic imaging at 7 T," *Magn. Reson. Med.*, vol. 88, no. 6, pp. 2339–2357, 2022, doi: 10.1002/mrm.29373.

[34] B. Guérin et al., "The ultimate signal-to-noise ratio in realistic body models," *Magn. Reson. Med.*, vol. 78, no. 5, pp. 1969–1980, 2017, doi: 10.1002/mrm.26564.

[35] C. Le Ster et al., "Magnetic field strength dependent SNR gain at the center of a spherical phantom and up to 11.7T," *Magn. Reson. Med.*, vol. 88, no. 5, pp. 2131–2138, 2022, doi: 10.1002/mrm.29391.

[36] F. Lam et al., "Ultrafast magnetic resonance spectroscopic imaging using SPICE with learned subspaces," *Magn. Reson. Med.*, vol. 83, no. 2, pp. 377–390, 2020, doi: 10.1002/mrm.27980.

[37] F. Lam et al., "High-resolution 1H-MRSI of the brain using SPICE: Data acquisition and image reconstruction," *Magn. Reson. Med.*, vol. 76, no. 4, pp. 1059–1070, 2016, doi: 10.1002/mrm.26019.

[38] R. Guo et al., "Simultaneous QSM and metabolic imaging of the brain using SPICE: Further improvements in data acquisition and processing," *Magn. Reson. Med.*, vol. 85, no. 2, pp. 970–977, 2021, doi: 10.1002/mrm.28459.

[39] R. Guo et al., "High-resolution label-free molecular imaging of brain tumor," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2021, pp. 3049–3052, doi: 10.1109/EMBC46164.2021.9630623.

[40] G. Hangel et al., "Ultra-high resolution brain metabolite mapping at 7 T by short-TR Hadamard-encoded FID-MRSI," *Neuroimage*, vol. 168, pp. 199–210, 2018, doi: 10.1016/j.neuroimage.2016.10.043.

[41] R. J. Ogg, P. B. Kingsley, and J. S. Taylor, "WET, a T1- and B1-insensitive water-suppression method for in vivo localized 1H NMR spectroscopy," *J. Magn. Reson. B.*, vol. 104, pp. 1–10, 1994, doi: 10.1006/jmrb.1994.1048.

[42] R. Guo et al., "Mapping intracellular NAD content in entire human brain using phosphorus-31 MR spectroscopic imaging at 7 Tesla," *Front Neurosci.*, vol. 18, 2024, Art. no. 1389111, doi: 10.3389/fnins.2024.1389111.

[43] Z. P. Liang, "Spatiotemporal imaging with partially separable functions," in *Proc. 4th IEEE Int. Symp. Biomed. Imag.: From Nano to Macro - Proc.*, 2007, pp. 988–991, doi: 10.1109/ISBI.2007.357020.

[44] Z. P. Liang and P. C. Lauterbur, "A generalized series approach to MR spectroscopic imaging," *IEEE Trans. Med. Imag.*, vol. 10, no. 2, pp. 132–137, 1991, doi: 10.1109/42.79470.

[45] R. Guo et al., "Simultaneous mapping of water diffusion coefficients and metabolite distributions of the brain using MR spectroscopic imaging without water suppression," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 3, pp. 962–969, Mar. 2023, doi: 10.1109/TBME.2022.3206584.

[46] Y. Li et al., "Improved estimation of myelin water fractions with learned parameter distributions," *Magn. Reson. Med.*, vol. 86, no. 5, pp. 2795–2809, 2021, doi: 10.1002/mrm.28889.

[47] O. Boldea and J. R. Magnus, "Maximum likelihood estimation of the multivariate normal mixture model," *J. Amer. Stat. Assoc.*, vol. 104, no. 488, pp. 1539–1549, 2009, doi: 10.1198/jasa.2009.tm08273.

[48] Q. Ning et al., "Spectral quantification for high-resolution MR spectroscopic imaging with spatiospectral constraints," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 5, pp. 1178–1186, May 2017, doi: 10.1109/TBME.2016.2594583.

[49] Y. Liu et al., "Improved low-rank filtering of magnetic resonance spectroscopic imaging data corrupted by noise and B0 field inhomogeneity," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 4, pp. 841–9, Apr. 2016, doi: 10.1109/TBME.2015.2476499.

[50] X. Peng et al., "Correction of field inhomogeneity effects on limited *k*-space MRSI data using anatomical constraints," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2010, pp. 883–886, doi: 10.1109/IEMBS.2010.5627873.

[51] M. Esmaeili et al., "Whole-slab 3D MR spectroscopic imaging of the human brain with spiral-out-in sampling at 7T," *J. Magn. Reson. Imag.*, vol. 53, no. 4, pp. 1237–1250, 2021, doi: 10.1002/jmri.27437.

[52] K. P. Pruessmann et al., "SENSE: Sensitivity encoding for fast MRI," *Magn. Reson. Med.*, vol. 42, no. 5, pp. 952–962, 1999, doi: 10.1002/(SICI)1522-2594(199911)42:5<952::AID-MRM16>3.0.CO;2-S.

[53] L. I. Sacolick et al., "B1 mapping by Bloch-Siegert shift," *Magn. Reson. Med.*, vol. 63, no. 5, pp. 1315–22, 2010, doi: 10.1002/mrm.22357.

[54] J. L. Kent et al., "Rapid 3D absolute B1+ mapping using a sandwiched train presaturated TurboFLASH sequence at 7 T for the brain and heart," *Magn. Reson. Med.*, vol. 89, no. 3, pp. 964–976, 2023, doi: 10.1002/mrm.29497.